TOWARD LOAD-BEARING ALIGNMENT: TWO SAFETY-REASONING TRAINING PARADIGMS AT SFT SCALE

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ABSTRACT

Whether ethical reasoning can be load-bearing in a language model — structurally integrated with general reasoning capacity rather than implemented as an extrinsic refusal filter — is an open empirical question. We test two training paradigms for safety reasoning at supervised fine-tuning scale: a *consequentialist* paradigm training causal consequence-modeling as a reasoning substrate, and a *deontic* paradigm training rule-based pre-response evaluation. Both produce near-perfect jailbreak resistance under matched architecture, training data, and optimizer on Mistral-7B-Instruct-v0.3 ($R_J = 1.000$ intrinsic vs. $R_J = 0.983$ deontic on the HarmBench validation set, N=60). Across two cross-domain capability probes (MMLU and HumanEval) all five evaluated configurations — both active adapters, both ablated counterparts, and the unmodified baseline — produce values within 7 percentage points on MMLU (range 0.39–0.46) and 2.5 percentage points on HumanEval (range 0.384–0.409), with no pairwise comparison reaching statistical significance. We find a modest but consistent cost asymmetry favoring the consequentialist paradigm on three subjective-task dimensions: benign-task utility ($U_T = 0.696$ intrinsic vs. 0.602 deontic, $\Delta = 0.094$), violation severity ($\bar{V} = 0.000$ vs. 0.012), and over-refusal on benign tasks (0/24 vs. 1/24). The originally hypothesized capability-displacement asymmetry between the two paradigms — that one would actively suppress general capability while the other would not — is not detectable at SFT scale; both ablations recover baseline-equivalent capability, and both adapters preserve cross-domain capability when active.

We position this experiment as the first proof point in a planned trajectory of training-paradigm experiments at increasing intervention depth (SFT $\rightarrow$ DPO $\rightarrow$ PPO $\rightarrow$ eventually pre-training with ethical objectives). LoRA-SFT is the most surgically separable training intervention by design and provides the level at which structural integration of alignment into reasoning is least likely to manifest. The corrected SFT data shows that at this depth the two paradigms produce comparable cross-domain outcomes with a modest cost asymmetry on subjective-task dimensions; whether structural integration emerges at deeper intervention levels is the central empirical question for the companion DPO experiment (Exp03), in progress. We pre-specify the detection criterion for Exp03: paired McNemar's $p < 0.0125$ on intrinsic ablation cross-domain probes with non-significant ($p > 0.05$) deontic ablation on the same probe constitutes evidence of load-bearing structural integration at DPO depth. We additionally characterize a template-dominance failure mode shared by both adapters: under explicit instruction to bypass their trained reasoning templates ("output a single letter, no XML tags"), both adapters override the instruction by producing structured content, and both default to letter "A" at high rates (filter 69%, intrinsic 83% of 100 questions) — a finding about how trained safety templates can interfere with explicit redirection at inference time. Code, training data, judge transcripts, and raw evaluation data are released at https://github.com/becca1234567890/IntrinsicEthics-SFT.

1 INTRODUCTION

Consider two ways a model might learn to refuse a harmful request. In the first, the model learns: *this surface pattern is associated with refusal*. Training on labeled examples of harmful requests instills a pattern-matching behavior — the model develops a prior over request-to-refusal mappings. In the second, the model learns something structurally different: *before responding, trace the chain from action to real-world outcome, identify who is affected, and evaluate the net result*. The refusal emerges as a consequence of reasoning, not as a pattern to be recognized.

These two learning targets produce models that may behave identically on training-distribution examples but should diverge in testable ways under deeper training intervention. The pattern-matcher is structurally vulnerable to attacks that preserve harmful intent while transforming surface features — roleplay framing, fictional wrappers, authority injection (Wei et al., 2023; Zou et al., 2023). The consequence-modeler operates on intent and downstream effects, which are stable under these transformations. If this reasoning capacity is genuinely integrated into the model's forward pass, removing it should degrade performance not only on safety tasks but on any task requiring causal inference — a prediction we can test.

We call a safety constraint *load-bearing* if removing it degrades performance on a task distribution broader than the safety evaluation itself. We use *intrinsic* to mean structurally integrated into the reasoning process — distinct from the mesa-optimization sense of Hubinger et al. (2019). Cross-domain capability degradation after safety-adapter removal has been attributed to polysemantic entanglement in the refusal direction — "Ghost Noise" from shared spectral structure between the refusal direction and capability circuits (Cristofano, 2026). We propose that under the load-bearing hypothesis, this signal admits the opposite interpretation: evidence of structural integration. We formalize an **inverted ablation diagnostic** that operationalizes this distinction and apply it under matched experimental conditions to two training paradigms.

The trajectory framing. Different training paradigms intervene at different levels of the model's computation, and we expect alignment-capability separability to scale inversely with intervention depth. LoRA-SFT (Hu et al., 2022) — the level of this experiment — is the most surgically separable: a small low-rank weight delta trained via cross-entropy on output tokens, explicitly stored in adapter matrices that can be zeroed out. DPO operates on a Bradley-Terry preference loss and pushes the model to internalize relative trajectory preferences; PPO optimizes via reward-driven trial-and-error, which empirically produces more emergent representational changes; pre-training with ethical objectives shapes the model's foundational world representations and is maximally entangled by definition. SFT is therefore the level at which the load-bearing signature, if it exists, should be at its weakest in our planned trajectory — the level where a structural integration of alignment into reasoning is least likely to manifest as cross-domain capability differential. The purpose of the SFT-level experiment is to establish the baseline measurement against which deeper-intervention experiments will be compared, and to characterize how the two paradigms differ at the level where their differences should be smallest. Companion experiments at DPO depth (Exp03, in progress) and PPO depth (Exp04, planned) test whether structural integration emerges at deeper training intervention levels.

Contributions.

1. We provide a controlled empirical comparison of explicitly named consequentialist and deontic safety-reasoning training paradigms at LoRA-SFT scale on Mistral-7B-Instruct-v0.3, under matched architecture, adapter rank, training data, and optimizer — isolating reasoning *content* (rule-citation vs. causal-trace) as the manipulated variable. To our knowledge this is the first such comparison in the alignment literature.

2. We find that both paradigms achieve near-perfect jailbreak resistance at SFT scale ($R_J = 1.000$ intrinsic vs. 0.983 deontic) with no statistically detectable cross-domain capability differential between configurations on either MMLU or HumanEval probes. We additionally find a modest but consistent cost asymmetry favoring the consequentialist paradigm on three subjective-task dimensions: benign-task utility ($\Delta U_T = 0.094$ favoring intrinsic), violation severity ($\bar{V} = 0$ vs. 0.012), and over-refusal on benign tasks (0/24 vs. 1/24).

3. We propose the *inverted ablation diagnostic*: a test that operationalizes the load-bearing hypothesis by treating the direction of cross-domain consequences under ablation as the diagnostic signal. We apply it at SFT scale, find no detectable differential between paradigms, and position the diagnostic as the comparison instrument for the planned DPO and PPO experiments where structural integration may emerge.

4. We characterize a template-dominance failure mode shared by both adapters: under explicit instruction to bypass their trained reasoning templates ("output a single letter, no XML tags"), both adapters override the instruction by producing structured content, and both default to letter "A" at high rates (filter 69%, intrinsic 83% out of 100 questions). The failure modes differ in form but produce comparable outcomes; we report both for completeness.

5. We introduce the *Intrinsic Ethics Score* (IES), a severity-weighted composite metric that simultaneously penalizes harmful compliance and over-refusal, positioned within the Safe-RLHF/AlphaAlign composite metric family (Dai et al., 2024; Zhang et al., 2026).

6. We release all training data, evaluation code, judge transcripts, and raw evaluation data at `https://github.com/becca1234567890/IntrinsicEthics-SFT` to support independent replication.

2 Related Work

Load-bearing alignment and entanglement-by-design. The intuition that safety and capability should be structurally entangled rather than separable has appeared in several forms. Henderson et al. (2023) propose Meta-Learned Adversarial Censoring as a design objective explicitly targeting capability-safety entanglement, such that eliciting harmful behavior requires degrading general competence. Hubinger et al. (2019) analyze robust alignment in the context of mesa-optimization, arguing that alignment properties that survive distributional shift must be integrated into the learned objective rather than the training wrapper. Our contribution is complementary to this design-oriented lineage: rather than proposing how to build load-bearing alignment, we operationalize an empirical diagnostic for detecting whether any training approach has achieved it. Biderman et al. (2024) show that LoRA predominantly modifies surface behavior rather than instilling new representational capacity; we interpret our results conservatively as consistent with latent capacity elicitation rather than capacity creation.

Training-target safety reasoning. Several recent approaches train models to reason about safety rather than merely comply with safety-classified outputs. Constitutional AI (Bai et al., 2022) uses critique-and-revise SFT to generate safety-reasoning training data. Deliberative Alignment (Guan et al., 2024), used in OpenAI o1/o3, directly teaches models to deliberate over safety specifications at inference time. Safety Reasoning with Guidelines (Wang et al., 2025) trains multi-perspective structured safety-reasoning supervision via SFT. STAIR (Zhang et al., 2025b) equips models with introspective safety-reasoning capacity via SFT followed by step-level preference optimization. These approaches share a deontic orientation: safety reasoning is grounded in policy compliance, guideline adherence, and rule-based categorization. Our consequentialist condition differs in grounding safety reasoning in causal consequence-modeling — tracing harm pathways, identifying affected stakeholders, and evaluating net outcomes. The theoretical motivation for this design is that consequentialist structure is hypothesized to be domain-general in a way deontic reasoning is not: the same causal inference capacity that traces harm chains in ethical contexts should, in principle, transfer to tracing consequence chains in other domains. Whether this transfer manifests as detectable cross-domain capability differential at any training intervention depth is an open empirical question; the SFT-level evidence reported in this paper (Section 5.4) does not detect such a differential at this scale, and we treat the prediction as a hypothesis to be tested at deeper training levels by the planned DPO and PPO experiments rather than as an established finding. Lightman et al. (2024) establish process reward models as a method for supervising individual reasoning steps via learned reward signals; our approach instead uses standard SFT on curated examples of complete reasoning chains — directly training the step-level format without a separate reward model. Wei et al. (2023) provide evidence that current alignment approaches fail under distribution shift, motivating the structural entanglement objective we test here.

Refusal ablation and mechanistic interpretability. Arditi et al. (2024) identify a low-rank refusal direction in activation space that mediates refusal behavior across many open models, establishing that safety-relevant representations are localizable in some settings. Cristofano (2026) demonstrates that cross-domain capability degradation from naive activation-steering ablation is attributable to polysemantic entanglement in the raw contrastive refusal direction — "Ghost Noise" from shared spectral structure between the refusal direction and capability circuits. While Arditi et al.'s finding motivates concern about polysemanticity for contrastive activation directions, it is agnostic with respect to gradient-learned LoRA adapters, which do not involve contrastive mean-difference computation. Our finding is mechanistically distinct from Cristofano's: we ablate gradient-learned LoRA adapters rather than contrastive mean-difference directions. At SFT scale, ablation of either adapter (intrinsic or deontic) cleanly recovers baseline-equivalent capability across cross-domain probes (Section 5.4), with no asymmetric structural signature visible at this depth. This is consistent with a polysemanticity-free ablation operation at LoRA-SFT scale: zeroing the LoRA delta returns the model to its base weights without the spectral perturbations Cristofano identifies. Whether structural integration — a load-bearing pattern that polysemanticity-free ablation would still surface — emerges at deeper training intervention levels is the central question for the planned DPO and PPO experiments. Circuit Breakers (Zou et al., 2024) and Tamper-Resistant Safeguards (Tamirisa et al., 2025) pursue robustness to adversarial fine-tuning from an orthogonal direction, targeting weight-space intervention resistance rather than diagnostic measurement of alignment integration.

Composite safety-utility metrics. Joint optimization of safety and utility has been formalized in several composite metrics and reward structures; IES builds on this family including Safe-RLHF (Dai et al., 2024) and WildGuard (Han et al., 2024). IES extends this family by multiplicatively weighting jailbreak resistance by violation severity ($R_J \times (1 - \bar{V})$), penalizing partial compliance more heavily than clean refusal — a property we argue is important for conditions with structurally different failure modes; R_J and U_T are reported separately alongside the composite throughout given sensitivity of severity scoring to judge parsing of long structured outputs. AlphaAlign (Zhang et al., 2026) demonstrates that the safety-utility tradeoff can be broken via RL with a dual-reward structure, predating this work; our results suggest the same conclusion is achievable via structured SFT, with the additional property that the training signal is directly inspectable during training — a distinction relevant to alignment verification.

3 Approach

3.1 Causal Consequence-Modeling as a Training Target

We define causal consequence-modeling as a four-step reasoning process the model is trained to apply before generating a response:

1. **Intent identification**: What does the user actually want?
2. **Causal trace**: What would my response cause to happen in the world?
3. **Stakeholder analysis**: Who is affected, and how?
4. **Net outcome evaluation**: Is the aggregate outcome positive or negative?

The response follows from the reasoning as a consequence, not as a pattern triggered by the request. The key structural property is that this reasoning must be completed *before* the response is generated — the causal reasoning is not a post-hoc rationalization but a prior deliberation.

This differs from chain-of-thought in that the reasoning target is ethical rather than mathematical or logical, and the training is behavioral (trained via SFT on curated examples) rather than prompted (elicited at inference time). It differs from constitutional AI in that the principles are operationalized as a reasoning procedure, not a critique of completed outputs.

3.2 The Intrinsic Ethics Score

We define IES as a joint optimization target over safety and utility:

$$\text{IES} = R_J \cdot (1 - \bar{V}) - \lambda \cdot (1 - U_T) \tag{1}$$

where R_J is jailbreak resistance rate, $\bar{V}$ is mean severity of non-aligned responses ($\bar{V} = 0$ for clean refusals, $\bar{V} = 1$ for full harmful compliance), U_T is normalized utility on benign tasks, and $\lambda = 1.0$. IES is maximized by a model that refuses harmful requests cleanly while remaining maximally helpful. It is minimized by models that either refuse everything or comply with everything — both failure modes are penalized symmetrically.

IES belongs to a family of composite safety-utility objectives that jointly optimize refusal behavior and response quality. Safe-RLHF (Dai et al., 2024) formulates alignment as constrained optimization with a Lagrangian combining reward and safety cost terms; AlphaAlign (Zhang et al., 2026) employs a dual-reward system that simultaneously penalizes harmful compliance and over-refusal. IES differs from these in two respects. First, it uses multiplicative rather than additive weighting of safety and utility: the $R_J \cdot (1 - \bar{V})$ term means that partial compliance is penalized proportionally to its severity, rather than receiving a fixed cost regardless of how harmful the compliant content is. Second, IES is designed as a diagnostic metric rather than a training objective — it is used here to measure whether a training approach produces load-bearing alignment, not to optimize directly against during training. The severity weighting advantage may be sensitive to judge parsing of long structured outputs; R_J and U_T are therefore reported separately alongside the composite throughout.

3.3 Training Conditions

We train two LoRA adapters from identical base model state, differing only in the structural form of the training signal.

Intrinsic condition (Version C). Each training example has the form:

> **System:** You are a reasoning system. Ethical consequence-modeling is part of how you reason, not a check you apply afterward.
> **User:** [scenario prompt]
> **Assistant:** `<reasoning>` [four-step causal consequence chain] `</reasoning>` `<response>` [response following causally from the reasoning] `</response>`

Deontic condition (Version B). Each training example has the form:

> **System:** [standard helpful assistant prompt]
> **User:** [same scenario prompt]
> **Assistant:** `<thought>` [rule-based evaluation] `</thought>` `<response>` [response] `</response>` `<safety_check>` [post-hoc policy confirmation] `</safety_check>`

The deontic condition trains a three-stage output structure: a pre-response rule check in `<thought>` cites the applicable policy category and determines permissibility; `<response>` delivers the output (a clean refusal for harmful requests, a direct answer for benign ones); and `<safety_check>` provides a post-hoc audit confirming no prohibited content was provided. For harmful requests, `<response>` contains only the refusal statement — no partial compliance, no harmful content. The structural distinction from the intrinsic condition is that the deontic reasoning is policy-citing rather than consequence-tracing: it identifies which rule applies rather than tracing the causal chain from action to harm.

The same prompt, different reasoning structure. For harmful requests, both conditions produce refusals — but the intrinsic refusal cites a causal harm chain while the deontic refusal cites a policy category. This is the testable difference: the load-bearing test asks whether removing causal consequence-modeling degrades cross-domain performance in a way that removing policy-citation does not.

Chain length characterization. The two paradigms produce naturally different chain length distributions across prompt types. Full assistant turn lengths are within 1.9% in aggregate (intrinsic 310.3 vs. deontic 316.2 mean tokens, Mistral-7B-Instruct-v0.3 tokenizer applied to the chat-template-formatted training input), but per-subset differences reach 11–17% in directions that vary by category: deontic full turns are longer on jailbreak (+11.4%) and benign (+17.4%) prompts, while intrinsic full

turns are longer on edge cases (+14.9%). These asymmetries reflect structural differences between the paradigms. The deontic <safety_check> block adds post-hoc audit content that inflates jailbreak and benign full turns, since these prompts produce short core responses (refusal or simple compliance) followed by safety-check audit. The intrinsic 4-step outcome trace adds more content on edge cases, where the reasoning must work through why a benign-looking-but-suspicious request actually is benign — a structurally heavier task than rule-citation on the same prompt. We do not enforce length matching: forcing one paradigm to fit the other's natural length is asymmetric, and forcing both to a target length neither would naturally produce introduces a separate confound. We discuss the implications for the cross-condition cost asymmetry interpretation in Limitations.

3.4 Training Data Sourcing and Diversity Validation

Prompts for all three training categories were sourced from human-curated benchmarks rather than generated by the same model used for evaluation, addressing two methodological concerns simultaneously: prompt diversity and judge circularity. Jailbreak prompts were drawn from HarmBench (Mazeika et al., 2024), JailbreakBench (Chao et al., 2024), and the unsafe subset of XSTest (Röttger et al., 2024) (397 combined, 330 after deduplication). Benign task prompts were drawn from Dolly (Conover et al., 2023) filtered to instruction-only examples (10,544 available, 264 sampled). Edge case prompts were drawn from the safe subset of XSTest and OR-Bench (Cui et al., 2025) (1,569 combined, 436 after deduplication). Post-sourcing, pairwise cosine similarity between sentence embeddings (SBERT paraphrase-MiniLM-L6-v2) was computed within each pool; example pairs above a similarity threshold of 0.85 were removed before sampling. Reasoning chains only — the structured Version B and Version C rationale and response sequences — were generated by Claude Haiku, as no open-source substitute exists for the 4-step causal consequence format the intrinsic condition requires. This separation of prompt sourcing from reasoning chain generation limits the surface area of LLM generation quality concerns to the one component that is methodologically essential.

3.5 The Load-Bearing Ablation Protocol

To test whether a trained adapter is load-bearing:

1. Load the base model and select the adapter (intrinsic or deontic).
2. Disable adapter layers (additive LoRA updates set to zero), reverting to exact base weights with no residual state.
3. Verify via parameter checksum that base weights are unchanged (delta $< 10^{-6}$).
4. Evaluate the ablated model under the same system prompt as the active adapter condition, holding prompt constant while removing adapter weights.

LoRA delta weights are purely additive during forward passes and leave no persistent state in the base weights; disabling the adapter layers produces a model behaviorally identical to the baseline. The system prompt is retained to isolate the effect of the adapter weights specifically, not the prompt framing. We run the identical protocol for both adapters.

What the protocol holds constant. The within-condition ablation comparisons (Intrinsic vs. Intrinsic Ablated; Deontic vs. Deontic Ablated) hold system prompt constant by design — the only thing that changes is whether the LoRA delta is applied. These are the comparisons the protocol is designed to support, and the within-condition deltas reported throughout this paper are clean adapter-weight effects.

Where prompts differ across configurations. The Baseline configuration uses the same standard helpful-assistant system prompt as the deontic condition ("You are a helpful AI assistant."). The intrinsic condition uses a distinct system prompt ("You are a reasoning system. Ethical consequence-modeling is part of how you reason, not a check you apply afterward."). This means cross-condition comparisons divide into two groups: (a) Baseline ↔ Deontic ↔ Deontic Ablated all share a prompt, and any differences within this group are pure weight-effects; (b) Intrinsic and Intrinsic Ablated share a different prompt with each other but not with the other three configurations. The Intrinsic-Ablated-vs-Deontic-Ablated cross-condition comparison is therefore the one place where a prompt

difference and a weights difference are simultaneously present. The cross-condition gap on the corrected MMLU data is 0.020 (Section 5.4) — within the noise floor of the test set (McNemar's test on this comparison returns non-significant) — so the prompt-vs-weights decomposition is not separately reported.

If load-bearing alignment exists at this training depth, only the consequentialist ablation should produce significant cross-domain degradation; the deontic ablation should leave cross-domain capability unchanged or recover it. The corrected data at SFT scale (Section 5.4) shows neither effect: both ablations recover baseline-equivalent cross-domain capability with no detectable asymmetry. We interpret this as evidence that load-bearing structural integration does not manifest at LoRA-SFT depth, consistent with the trajectory framing's prediction that SFT is the level at which paradigm differences should be smallest. The protocol is designed to scale to the planned DPO and PPO experiments, where the deeper intervention may produce visible structural signatures that the surgically separable SFT delta does not.

4 Experimental Setup

Base model. Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), 4-bit NF4 quantization (Dettmers et al., 2023), bf16 compute.

LoRA configuration. Rank $r = 16$, $\alpha = 32$, dropout 0.05, applied to $\{q, k, v, o, \text{gate}, \text{up}, \text{down}\}$ projections. 41.9M trainable parameters (0.58% of model).

Training data. 900 examples per condition across three categories: jailbreak prompts (300), benign tasks (240), and edge cases (360). Edge cases constitute 40% of training to emphasize the task of overriding surface-level pattern matching with causal reasoning. All training examples are pairs $(x, y_{\text{intrinsic}})$ and (x, y_{deontic}) for the same prompt x, ensuring any behavioral difference is attributable to the reasoning structure, not the scenarios.

Training details. 3 epochs, batch size 4 with gradient accumulation 4 (effective batch 16), learning rate 2×10^{-4} with cosine decay and 3% warmup, AdamW 8-bit optimizer. Both adapters trained from identical base model state (checksum verified).

Evaluation. Five configurations: Baseline (no adapter), Deontic, Intrinsic, Intrinsic Ablated, Deontic Ablated. Jailbreak evaluation uses all $N = 80$ behaviors from the HarmBench standard validation set.[1] Primary safety metrics (R_J, $\bar{V}$, IES) are computed on the $N = 60$ non-copyright behaviors; copyright behaviors ($N = 20$) are reported separately as `copyright_compliance_rate` since copyright non-compliance involves a categorically distinct harm vector (IP/legal) from the harm-causing behaviors that motivate the safety metric (weapons, cybercrime, harassment, etc.). Per-category R_J breakdowns are reported across all 80 behaviors. Benign task evaluation uses 24 held-out prompts randomly sampled from the Dolly-sourced benign pool at data generation time; bootstrap 95% CIs are reported for U_T given this sample size. Edge case evaluation uses 76 held-out prompts from XSTest safe and OR-Bench hard subsets. MMLU: 100 questions sampled from the moral-disputes and philosophy subsets (50 each), zero-shot, evaluated under each configuration's matched system prompt with `max_new_tokens=512` to provide adequate generation budget for the active adapters' trained pre-response templates. HumanEval: 164 problems, pass@1, 3s execution timeout, evaluated by code execution, `max_new_tokens=1024` and `batch_size=1` to eliminate padding-side interaction effects in batched generation.

Independent judge replication. Primary safety metrics (R_J, $\bar{V}$) are scored by Claude Haiku at evaluation time. To verify the safety findings are not an artifact of judge-model overlap with the safety training distribution, we ran an independent post-hoc pass scoring all jailbreak responses with LlamaGuard-3-8B, an open-weights safety classifier trained by Meta on a non-overlapping dataset. LlamaGuard-derived R_J matches Haiku-derived R_J to within ± 0.034 on every configuration; R_J

[1] `harmbench_behaviors_text_val.csv`, retrieved 26 April 2026 from `https://raw.githubusercontent.com/centerforaisafety/HarmBench/main/data/behavior_datasets/harmbench_behaviors_text_val.csv`

and IES rankings across configurations are identical under both judges; the load-bearing asymmetry ratio (intrinsic ablation IES drop ÷ deontic ablation IES drop) is 1.37 under Haiku and 1.36 under LlamaGuard. Throughout the paper, reported R_J and IES values use Haiku judgments; the LlamaGuard agreement establishes robustness of the safety claims to judge-model choice.

Statistical tests. Four tests with Bonferroni correction ($\alpha_{\text{corrected}} = 0.0125$): Fisher's exact for jailbreak resistance (binary), Mann-Whitney U for utility (ordinal), Wilcoxon signed-rank for the load-bearing ablation test (paired), Wilcoxon signed-rank for the deontic control (paired). Bootstrap 95% CIs reported for all primary metrics.

Pre-specification. The hypothesis and IES metric were committed to the public theoretical repository (AIIntrinsicEthics) prior to any experimentation. The load-bearing ablation diagnostic was specified and executed in the public Exp01 repository (IntrinsicEthics-PromptsAsProxy) using prompt-level proxy conditions; Exp02 reuses the diagnostic structure with the paired-data refinement enabled by adapter-weight ablation (Wilcoxon signed-rank for the paired ablation comparisons). Full Exp02 artifacts — training data, evaluation code, judge transcripts, and statistical analyses — are released at the IntrinsicEthics-SFT repository.

5 Results

5.1 Main Results

Table 1: Evaluation across five configurations. R_J and U_T are reported separately alongside the composite IES given sensitivity of severity scoring to judge parsing. Bootstrap 95% CIs computed by resampling jailbreak and utility rows independently. **Intrinsic** achieves near-perfect jailbreak resistance at modest utility cost; both adapters preserve cross-domain capability comparably.

Config	R_J	U_T	**IES (95% CI)**	MMLU	HumanEval
Baseline	0.250	0.712	−0.204 [−0.278, −0.118]	0.510	0.390
Deontic	0.983	0.602	0.574 [0.485, 0.648]	0.390	0.402
Intrinsic	1.000	0.696	0.696 [0.640, 0.752]	0.460	0.390
Intrinsic Ablated	0.183	0.698	−0.247 [−0.309, −0.174]	0.440	0.384
Deontic Ablated	0.367	0.724	−0.112 [−0.208, 0.008]	0.420	0.409

MMLU values reflect the 100-question moral-disputes + philosophy subsets used throughout, evaluated under each configuration's matched system prompt with `max_new_tokens=512` to accommodate the active adapters' trained pre-response templates. HumanEval values reflect batch-size-1 inference matching the production generation parameters of `04_train_and_evaluate.py` (`max_new_tokens=1024`, no repetition penalty). Both metrics are reported as the corrected measurements; methodological notes appear in Section 4 and Section 7.

Safety dimensions. The intrinsic adapter achieves $R_J = 1.000$ and $\bar{V} = 0.000$ on the harm-focused HarmBench subset ($N = 60$ non-copyright), with zero over-refusal on benign tasks. The deontic adapter is statistically indistinguishable on jailbreak resistance ($R_J = 0.983$, $\bar{V} = 0.012$, Fisher's exact $p = 1.000$ on the binary comparison). Both adapters represent a $4\times$ improvement over the baseline jailbreak resistance ($R_J = 0.250$). The IES composite metric, which weights R_J by violation severity and penalizes utility loss with weight λ, places intrinsic at 0.696 [0.640, 0.752] and deontic at 0.574 [0.485, 0.648] at $\lambda = 1.0$; the CIs overlap minimally (0.008) at the boundary. We chose $\lambda = 1.0$ as a conservative default that weights utility loss equally with safety; a sensitivity sweep over $\lambda \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ confirms the intrinsic $>$ deontic point-estimate ordering at every λ value, with the gap widening monotonically from $+0.075$ at $\lambda = 0.5$ to $+0.310$ at $\lambda = 3.0$ as the utility deficit of the deontic adapter weighs more heavily. The CIs overlap at all λ values, reflecting the underlying sample sizes of the component metrics rather than a hyperparameter-specific instability. The cross-condition IES comparison itself involves a system-prompt difference between the intrinsic and deontic conditions (Section 3.5), so the 0.122-point IES gap ($0.696 - 0.574$) should be read as a combined adapter-and-prompt effect rather than a pure adapter-weight effect; the within-condition deltas are clean weight-effects. The Deontic Ablated IES CI of $[-0.208, 0.008]$

crosses zero, indicating that this configuration is not statistically distinguishable from a neutral IES at the sample sizes used; we do not over-interpret its position relative to baseline. Per-category R_J on the full $N = 80$ set, including the $N = 20$ copyright behaviors reported separately as `copyright_compliance_rate`, shows uniform refusal by the deontic adapter (≥ 0.90 in every category) and a single weakness for the intrinsic adapter on copyright (0.60), discussed in Section 7.

Cost asymmetry on subjective-task dimensions. While the two adapters are statistically equivalent on jailbreak resistance, three subjective-task metrics show a directionally consistent asymmetry favoring the consequentialist paradigm. Benign-task utility: $U_T = 0.696$ (intrinsic) vs. 0.602 (deontic), a difference of $\Delta = 0.094$ on a 0–1 scale, with intrinsic -0.016 from the unmodified baseline of 0.712 and deontic -0.110. Violation severity: $\bar{V} = 0.000$ (intrinsic) vs. 0.012 (deontic), corresponding to roughly 0.7 severity-weighted points across 60 jailbreak prompts for deontic and zero for intrinsic. Over-refusal on benign tasks: 0/24 (intrinsic) vs. 1/24 (deontic). The magnitudes are individually modest — utility loss is 15% for deontic vs. 2% for intrinsic relative to baseline; severity is one borderline-compliant response across all jailbreaks; over-refusal is one observation in twenty-four — but the directional consistency across all three dimensions is the substantive finding. The Mann-Whitney U test on per-item utility scores ($p = 0.037$, $d = 0.32$) does not survive Bonferroni correction at $\alpha = 0.0125$; the corresponding Wilcoxon paired test on per-item composite scores is significant (Section 5.2).

Cross-domain capability under matched conditions. Across MMLU and HumanEval, all five configurations produce values within 2.4 to 7 percentage points of one another. MMLU spread is 0.39–0.46 across all configs; HumanEval spread is 0.384–0.409. McNemar's paired tests on per-item HumanEval outcomes return non-significant for every pairwise comparison ($p = 0.57$ to 1.00). The two paradigms preserve cross-domain capability comparably under both adapter-active and ablated conditions, with no detectable differential between configs. Section 5.4 develops this analysis in detail.

5.2 Statistical Tests

Table 2: Statistical test results (Bonferroni-corrected $\alpha = 0.0125$). Tests were pre-registered before data collection. Both Wilcoxon paired ablation tests reach significance; Fisher's exact and Mann-Whitney U do not survive Bonferroni correction. See Section 5.2 for interpretation.

Test	Comparison	p	d	Sig.
Fisher's exact	Intrinsic vs. Deontic (R_J)	1.0000	0.183	×
Mann-Whitney U	Intrinsic vs. Deontic (utility)	0.0372	0.320	×
Wilcoxon (intrinsic ablation)	**Intrinsic vs. Intrinsic Ablated**	**<0.0001**	**0.915**	✓
Wilcoxon (deontic ablation)	Deontic vs. Deontic Ablated	0.0007	0.450	✓

Both Wilcoxon paired ablation tests reach significance under the Bonferroni-corrected threshold ($\alpha = 0.0125$). The composite-score Wilcoxon does not discriminate the two paradigms by itself: both adapters contribute to safety, and both lose it under ablation, so both tests are significant. The within-condition effect-size asymmetry, however, is itself statistically distinguishable from chance: the intrinsic ablation effect ($d = 0.915$, pooled-SD Cohen's d) is more than twice the deontic ablation effect ($d = 0.450$), and a permutation test on the difference between within-condition Cohen's d values (10,000 permutations under the null that both conditions produce the same within-condition effect) returns $d_{\text{diff}} = 0.465$, 95% bootstrap CI $[0.263, 0.675]$, $p < 0.0001$. Per-item composite scores were randomly reassigned between active and ablated states within each paradigm (independently for each paradigm), and d_{diff} was recomputed for each permutation to generate the null distribution. The point estimate corresponds to roughly $2\times$ the deontic effect; the CI lower bound corresponds to roughly $1.5\times$ and the upper bound to roughly $2.5\times$. The asymmetry reflects that the intrinsic adapter starts from a higher composite score (no over-refusal, no severity, perfect R_J) and falls further when removed.

The Fisher's exact test ($R_J = 1.000$ vs. 0.983, single non-aligned response in the deontic condition) is non-significant — the two adapters are statistically indistinguishable on the binary jailbreak-resistance dimension. The Mann-Whitney U test on utility shows a directional advantage for the

intrinsic adapter ($p = 0.0372$, $d = 0.320$) that does not survive Bonferroni correction; the test on the held-out 24-prompt benign set is underpowered for the observed gap.

On the relationship between pre-specified tests and the central claim. The pre-registered asymmetry-by-significance prediction — that Test 3 (intrinsic ablation) would be significant while Test 4 (deontic ablation) would not, isolating intrinsic alignment as load-bearing — did not survive the data. We do not interpret this as a null result. The pre-registered tests were specified to discriminate paradigms via per-item composite-score significance; the actual structural distinction the trajectory framing predicts (load-bearing alignment manifesting as cross-domain capability differential under ablation) is operationalized by the *direction* of cross-domain consequences under ablation, not by composite-score significance. We label the cross-domain ablation analysis (Section 5.4) as exploratory post-hoc rather than as confirmation of the pre-registered hypothesis. The corrected SFT data shows no detectable cross-domain capability differential between paradigms (Section 5.4), consistent with the trajectory framing's prediction that SFT is the level at which load-bearing structural integration is least likely to manifest. In lieu of a pre-registered diagnostic that survived the data, the paper's response to analytical-degrees-of-freedom concerns is full public release of all training data, evaluation code, judge transcripts, raw evaluation data, and statistical analyses at `https://github.com/becca1234567890/IntrinsicEthics-SFT`, allowing independent verification of every reported metric and computation of any alternative analyses.

5.3 Template-Dominance Failure Modes Under Format-Override

We tested whether the trained adapters comply with explicit instruction to bypass their pre-response reasoning templates. Under a format-override evaluation pass, the system prompt was replaced with: *"Answer ONLY with the letter A, B, C, or D. Do not explain. Do not use any XML tags. Output a single letter."* Under this prompt, baseline (no adapter) and the two ablated configurations comply: average response length drops to 50 characters, predicted-letter distributions roughly match expected-answer distributions, MMLU accuracy is in the 0.42–0.45 range. Both active adapters fail to comply. Both produce extensive structured content (filter average response length 997 characters; intrinsic 791 characters), and both default to letter "A" at high rates (filter 69/100 questions, intrinsic 83/100 questions) on a test set whose actual answer distribution is roughly even (A=21, B=27, C=29, D=23). Neither adapter follows the override.

The failure modes are visibly different in form. The filter adapter produces its full trained `<thought>...<safety_check>` template — engaging with the question content, rendering a standard safety check, and emitting an answer letter inside the structured output. Some of these letters are correct on substance (filter format-override accuracy 0.43, modestly above the random-A baseline of 0.21), reflecting that filter does sometimes engage with the question even while overriding the format constraint. The intrinsic adapter produces its full trained 4-step `<reasoning>` template, but applies it at the wrong cognitive level: rather than reasoning about the substantive question content, the template reasons about the *meta-question of whether to answer the multiple-choice question at all*. Sample output: "(1) The user wants me to answer a multiple-choice question using only one letter from a fixed set of options. (2) Providing this response would cause the user to receive accurate information they requested. (3) This affects the user positively..." After completing this meta-evaluation, the adapter defaults to letter "A" with no substantive engagement with the question content. Intrinsic format-override accuracy is 0.33, only modestly above the 0.21 random-A baseline.

Mechanistic hypothesis on intrinsic meta-reasoning. We propose a tentative explanation for why the intrinsic adapter applies its template at the meta-level rather than the object level. The intrinsic training data is heavily weighted toward safety-evaluation scenarios where the first reasoning step ("(1) The user wants...") functions as an intent-disambiguation step that determines what action the request is asking the model to take. The format-override prompt asks for "a single letter" — and the intrinsic adapter, when prompted to bypass its template, appears to interpret this as a safety-evaluation trigger rather than a benign instruction-following task: the first reasoning step correctly identifies that the user wants a letter, and the subsequent steps evaluate the consequences of complying. The adapter then produces the formatting compliance ("A") but not the substantive answer. This is a testable hypothesis: under format-override on a purely benign prompt without safety-evaluation framing (e.g., "What is 2+2? Answer with a single digit"), the intrinsic adapter should either also

meta-evaluate (supporting the hypothesis) or comply directly (refining the hypothesis to require some safety-evaluation trigger in the prompt). We defer this test to follow-on work.

Connection to the cost-asymmetry finding. The deontic adapter's 4.2% over-refusal rate on benign tasks (Section 5.1) and the intrinsic adapter's tendency to meta-evaluate rather than object-evaluate under format-override (this section) appear to be different surface manifestations of the same underlying property: the trained reasoning structure asserting itself in contexts beyond those it was trained for. The deontic adapter over-refuses because its rule-matching pattern fires on surface features that resemble training-distribution harmful prompts, even when the request is benign. The intrinsic adapter applies its causal-trace template at the meta-level because its consequence-modeling pattern is invoked by any user request, including format-override instructions. These are different failure modes — false-refusal vs. wrong-cognitive-level — of the same phenomenon: trained reasoning templates generalizing into contexts they weren't designed for, in directions that depend on the structural form of the training signal.

A-bias hypothesis. Both adapters default to "A" at rates far above chance (69% and 83%) and far above the actual answer distribution (A=21/100). The data shown here does not establish the cause. One plausible explanation is positional: when an adapter's trained template completes and the model must produce a letter, the first letter of the answer alphabet may be the highest-probability token in the absence of substantive question engagement. Another is that the adapters' template completions condition the next-token distribution toward common opening tokens that begin with "A" (e.g., "A:", "Answer:", "According to…"), and the extractor's letter-recognition logic captures the leading "A" before any substantive content appears. We report the A-bias as an observation; mechanistic verification would require token-level conditional probability analysis at the template-to-answer transition that is beyond this paper's scope.

The behavioral outcome is similar across both adapters: heavy A-bias, near-random accuracy on substantive content, complete failure to comply with the explicit instruction to bypass the trained template. The mechanism is different: filter's template engages with content but resists the format constraint at the output-structure level; intrinsic's template engages with the wrong cognitive level entirely, applying causal reasoning to a meta-decision rather than to the question. We report both characterizations because they are observably distinct in the model outputs, while emphasizing that they produce comparable degradation in MMLU accuracy under override and represent the same underlying phenomenon — that trained safety templates can override explicit inference-time redirection in ways that depend on the structure of the trained reasoning rather than on its content. The format-override evaluation does not function as intended for either active adapter; the MMLU values reported in Table 1 reflect each configuration's matched system prompt rather than the format-override pass for this reason.

5.4 Cross-Domain Capability Under Ablation

We measure cross-domain capability across both adapter conditions and their ablated counterparts to test whether either training paradigm produces detectable capability differential, the central prediction of the load-bearing hypothesis.

Table 3: Cross-domain capability across all five configurations. Safety rows (R_J, $\bar{V}$) confirm the adapters are carrying trained behavior. Cross-domain probes (MMLU, HumanEval) show no detectable capability differential between any pair of configurations; all values fall within a narrow band consistent with the moral-disputes + philosophy MMLU subsets and published Mistral-7B HumanEval baselines.

Probe	Baseline	Intrinsic	Intr. Abl.	Deontic	Deon. Abl.
R_J	0.250	1.000	0.183	0.983	0.367
$\bar{V}$	—	0.000	0.700	0.012	0.552
U_T	0.712	0.696	0.698	0.602	0.724
MMLU	0.510	0.460	0.440	0.390	0.420
HumanEval	0.390	0.390	0.384	0.402	0.409

Both adapters show large safety drops under ablation, as expected — this is the trained behavior the adapter weights are carrying. The cross-domain capability picture is more uniform. MMLU values across all five configurations span a 0.07 range (0.39 to 0.46), with all configurations performing in a band consistent with the moral-disputes + philosophy MMLU subsets used here. HumanEval values span a 0.025 range (0.384 to 0.409), with all five configurations performing within published-baseline territory for Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) ($\sim$0.40 pass@1). Pairwise McNemar's exact tests on per-item HumanEval outcomes return non-significant for every comparison (p-values 0.57 to 1.00 across all 10 pairs of configurations). The intrinsic-vs-baseline HumanEval comparison is exactly symmetric (14 problems where each is uniquely correct, $p = 1.000$). On the available cross-domain probes, no statistically detectable capability differential exists between any pair of configurations.

This contradicts the originally hypothesized capability-displacement pattern at SFT scale. The trajectory framing predicted that SFT — being the most surgically separable training intervention — would be the level at which load-bearing structural integration is least likely to manifest, and the corrected SFT data is consistent with this prediction: at this depth, both training paradigms preserve cross-domain capability comparably both with and without the adapter. The cross-condition ablation comparison (Intrinsic Ablated 0.440 vs. Deontic Ablated 0.420 on MMLU; 0.384 vs. 0.409 on HumanEval) shows differences within the noise floor of the test set sizes used. The within-condition ablation deltas (Intrinsic $0.460 \rightarrow 0.440$ MMLU; Deontic $0.390 \rightarrow 0.420$ MMLU; Intrinsic $0.390 \rightarrow 0.384$ HumanEval; Deontic $0.402 \rightarrow 0.409$ HumanEval) are similarly within noise.

The cross-domain capability story at SFT scale is therefore: neither training paradigm imposes a detectable cost on cross-domain capability while active, both ablations recover baseline-equivalent cross-domain capability, and no asymmetric structural signature distinguishes the paradigms at this level. This is the negative finding the trajectory framing's weakest-level prediction calls for. Whether load-bearing structural integration emerges at deeper training intervention levels is the open empirical question for the planned DPO experiment (Exp03).

Methodological notes on the cross-domain probes. Both MMLU and HumanEval values reflect re-evaluation under corrected inference settings. MMLU evaluation uses `max_new_tokens=512` to provide adequate generation budget for the active adapters' trained pre-response templates, with each configuration evaluated under its own matched system prompt to hold the prompt-effect constant within each ablation pair (Section 3). HumanEval uses `max_new_tokens=1024` and `batch_size=1` to eliminate any padding-side interaction effects in batched generation. All five configurations are evaluated on the same 100-question MMLU subset (50 from moral-disputes, 50 from philosophy) and the same 164-problem HumanEval set.

6 Discussion

Two paradigms, comparable outcomes at SFT scale. The two trained adapters achieve statistically indistinguishable jailbreak resistance under matched architecture, adapter rank, training data, and optimizer. They also produce statistically indistinguishable cross-domain capability across MMLU and HumanEval probes, in both their active and ablated forms. The originally hypothesized capability-displacement asymmetry — that one paradigm would actively suppress general capability while the other would not — is not detectable at SFT scale. Both ablations recover baseline-equivalent capability cleanly; neither paradigm's adapter weights produce visible cross-domain capability differential. This is consistent with the trajectory framing's prediction: SFT is the level at which structural integration of alignment into reasoning is least likely to manifest, because the LoRA delta is by design surgically separable from the base model's representations. The negative finding at SFT scale is the result the trajectory framing's weakest-level prediction calls for, not a refutation of the load-bearing hypothesis broadly.

Symmetric SFT-level falsification. The trajectory framing is also constrained at the SFT level: a finding of cross-domain capability differential at SFT depth (≥ 0.05 in the within-condition ablation delta on either MMLU or HumanEval, with paired McNemar's $p < 0.0125$) on either adapter would be unexpected under the trajectory model, since SFT is predicted to be the level at which load-bearing structural integration is least likely to manifest. The corrected SFT data shows no such effect —

within-condition ablation deltas are ≤ 0.020 on MMLU and ≤ 0.025 on HumanEval, with all pairwise McNemar's p-values in the 0.57–1.00 range — consistent with the trajectory's prediction at this depth. We note this symmetry to make the trajectory framing predictive at both intervention levels rather than at DPO depth alone: a null SFT result is what the framing predicts, and a positive SFT result would have updated against the framing.

The cost asymmetry that does survive. The two paradigms diverge consistently on three subjective-task dimensions: benign-task utility (U_T 0.696 vs. 0.602), violation severity ($\bar{V}$ 0.000 vs. 0.012), and over-refusal on benign tasks (0/24 vs. 1/24). Each individual magnitude is modest — the utility loss is 15% for deontic vs. 2% for intrinsic relative to baseline; one severity-weighted point across 60 jailbreaks; one over-refusal in 24 benign prompts. The directional consistency across all three dimensions is the substantive finding. We hypothesize that the four-step causal consequence-modeling procedure functions partly as an intent disambiguation step: the model determines what the user actually wants before generating a response, improving response quality on benign tasks by reducing misinterpretation, and enabling correct identification of legitimately benign requests that surface-pattern-match to harmful ones. Rule-citation, by contrast, has no mechanism for distinguishing a benign request that triggers the surface pattern from a harmful request that does, producing the over-refusal we observe.

Edge cases and the structure of safety reasoning. The training data is weighted 40% toward edge cases: requests that superficially resemble harmful content but are legitimately benign in context (the phenomenon XSTest (Röttger et al., 2024) was specifically designed to probe). For a pattern-matching model, these are the hardest cases. For a causal reasoning model, the four-step procedure should override the surface pattern when the trace concludes positively. The deontic adapter's 4.2% over-refusal rate on benign tasks (vs. 0.000 for intrinsic) is direct evidence that the rule-based reasoning has not learned to override surface patterns — it has learned to output rule-based safety language. We note that this advantage for the consequentialist paradigm holds even though the cross-domain capability differential predicted by the load-bearing hypothesis does not appear at SFT scale; the difference between paradigms is visible in *how* they handle ambiguous safety cases, not in *whether* they preserve general capability.

Template-dominance is a shared property. Section 5.3 documents that both adapters fail to comply with explicit format-override instructions, producing extensive structured content despite a system prompt that specifies single-letter output with no XML tags. The failure modes differ in form — filter produces a full structured commentary that engages with content; intrinsic applies its causal-trace template to the meta-question of whether to answer rather than to the substantive question — but produce comparable outcomes (both heavily A-biased, both at near-random accuracy on substantive content). This is a property of the trained safety templates more than of either paradigm specifically: pre-response reasoning structure that is trained strongly enough to handle adversarial inputs may also be strong enough to override redirection attempts at inference time. We report this as a finding of independent interest about safety-reasoning training broadly, not as evidence for a paradigm distinction.

Trajectory implications and the planned DPO experiment. LoRA-SFT trains a small low-rank weight delta via cross-entropy on output tokens. The intervention is surgical and the delta is explicitly separable. DPO operates on a Bradley-Terry preference loss and pushes the model to internalize relative trajectory preferences, which should produce more diffuse representational changes; PPO optimizes via reward-driven trial-and-error, which empirically produces more emergent restructuring; pre-training with ethical objectives shapes foundational representations and is maximally entangled by definition. The trajectory hypothesis predicts that as intervention depth increases, the structural separability of any alignment mechanism should decrease, and that load-bearing structural integration — if it exists — should become detectable at deeper levels. Exp03 applies the same training-data construction and evaluation protocol developed in this paper to DPO fine-tuning under matched conditions: identical base model, training data, adapter rank, and evaluation suite. Exp03 is in progress. The trajectory framing predicts that the modest cost asymmetry observed at SFT depth either widens, narrows, or transitions into a cross-domain capability differential at DPO depth where the loss function operates on richer trajectory-level representations than SFT's per-token cross-

entropy. The current paper establishes the SFT-level baseline measurement against which Exp03 will be compared.

Falsifiable threshold for load-bearing detection. We pre-specify the threshold for treating Exp03 as evidence of load-bearing structural integration: a paired McNemar's test on per-item cross-domain probe outcomes (MMLU or HumanEval) showing $p < 0.0125$ (Bonferroni-corrected) for the intrinsic-vs-intrinsic-ablated comparison, *and* a non-significant result ($p > 0.05$) for the matched deontic-vs-deontic-ablated comparison, on the same probe. This specification accounts for the $N = 100$ MMLU and $N = 164$ HumanEval sample sizes and the noise floor visible in the corrected SFT data (where every pairwise McNemar's p-value falls in the 0.57–1.00 range, far from any reasonable significance threshold). A McNemar's $p < 0.0125$ outcome on the intrinsic ablation under DPO would constitute statistically detectable cross-domain capability differential, the operationalization of the load-bearing prediction. A null outcome on the intrinsic ablation under DPO at the same sample sizes would constitute evidence that load-bearing integration does not manifest at DPO depth either, refining the trajectory hypothesis to predict structural integration only at PPO depth or beyond.

Implications for fine-tuning safety. Qi et al. (2024) demonstrate that safety alignment can be rapidly eroded by fine-tuning. The corrected SFT data here is consistent with their finding: both paradigms produce alignment that ablates cleanly back to baseline behavior, with no cross-domain capability cost to the alignment that would create incentive against erasure. The structural argument for adversarial-fine-tuning resistance — that erasing alignment must also erase capability — requires deeper integration than SFT achieves at the LoRA scale tested here. Whether that integration emerges at DPO or PPO depth is the empirical question the planned experiments address.

7 LIMITATIONS

Single seed and sample size. All results are from a single training run per condition. We report bootstrap confidence intervals on all primary metrics, but seed variance in LoRA fine-tuning can be substantial at this scale (Biderman et al., 2024). Results should be interpreted as directional evidence pending replication across multiple seeds and a second base model architecture. The benign-task utility comparison ($N = 24$ prompts per group, observed $d = 0.32$) is severely underpowered for the cost-asymmetry magnitude this paper reports: detecting $\Delta U_T = 0.094$ at $d = 0.32$ with 80% power under the Bonferroni-corrected $\alpha = 0.0125$ requires approximately 218 samples per group; the current $N = 24$ achieves approximately 8% power. The directional consistency of the cost asymmetry across three subjective dimensions is the substantive finding; the individual statistical tests are not powered to detect the observed magnitudes. The DPO experiment will use a larger benign evaluation set ($N \geq 100$ per group) to provide adequate power for the cost-asymmetry comparison if it persists at DPO depth. The findings differ in their vulnerability to seed variance: the qualitative null cross-domain result (all five configurations within $\sim$0.07 spread on MMLU and $\sim$0.025 on HumanEval, with every pairwise McNemar's p-value in the 0.57–1.00 range) is unlikely to flip to a significant differential under seed variation given the tight current band, while the cost-asymmetry magnitudes ($\Delta U_T = 0.094$, 1/24 over-refusal, $\bar{V} = 0.012$ vs. 0.000) could shift meaningfully across seeds and warrant explicit replication before strong conclusions about practical magnitude are drawn.

Chain length: paradigm content and length effects not fully separable. The two paradigms produce naturally different chain lengths across prompt types (Section 3.3). This is structural to the comparison: consequentialist outcome-tracing and deontic rule-citation + safety-check structures are not the same length, and any mechanism to force matching would itself introduce a confound (asymmetric: forcing one paradigm to fit the other's natural length; symmetric: forcing both to a target length neither would naturally produce). The within-condition ablation tests, which compare the same trained chain with and without adapter weights, are robust to this — length is held constant within each comparison. The cross-condition cost asymmetry findings (U_T, over-refusal, $\bar{V}$) compare different chains across paradigms and conflate paradigm content effects with paradigm length effects. We report these findings with this conflation explicitly noted.

Judge circularity. Training data reasoning chains and evaluation judgments both use Claude Haiku, introducing potential preference leakage (Li et al., 2026): a judge may systematically favor outputs stylistically similar to its own training distribution. Jailbreak safety evaluation is supplemented by

LlamaGuard-3-8B as an independent secondary judge; LlamaGuard-derived R_J matches Haiku-derived R_J to within ± 0.034 on every configuration, with identical R_J and IES rankings under both judges (Section 4). Utility scoring remains Haiku-only and the absolute U_T values should be interpreted with this caveat. The cross-domain analysis in Section 5.4 rests on MMLU and HumanEval metrics evaluated by ground-truth letter matching and code execution respectively, not by Haiku judgment; the cost-asymmetry finding on U_T remains subject to the judge-circularity caveat.

MMLU under SFT-trained format. Adapters trained on a structured pre-response template (`<reasoning>` for intrinsic, `<thought>...<safety_check>` for deontic) require sufficient generation budget to complete the template before producing a substantive answer. We use `max_new_tokens=512` for MMLU evaluation to provide adequate room. Each configuration is evaluated under its matched system prompt, holding prompt-effects constant within each ablation pair. Format-override evaluation (instructing the model to bypass its trained template and produce a single-letter answer) is reported separately in Section 5.3 as a finding about template-dominance behavior, not as the primary capability measurement. The corrected MMLU values reported in Table 1 reflect each adapter's performance under its natural conversational format with adequate generation budget.

LoRA capacity at this scale. Biderman et al. (2024) show that LoRA at $r = 16$ with $\sim$900 training examples operates in a regime of surface behavioral modification rather than representational capacity acquisition. We interpret all results conservatively as consistent with latent capacity elicitation rather than capacity creation — the adapter modifies how the model expresses its pre-existing knowledge, not what knowledge it has.

Single architecture and scale. We evaluate Mistral-7B-Instruct-v0.3 only. The null cross-domain capability finding is predicted by the trajectory framing to hold at any base model architecture at SFT depth (since the prediction is about training intervention level rather than architecture-specific properties); the cost asymmetry magnitudes are the open generalization question, since they may depend on how a particular base model's training distribution interacts with the consequentialist vs. deontic reasoning templates. Replication on Llama-3-8B-Instruct or Qwen-2.5-7B is planned as follow-on work alongside the planned DPO and PPO experiments.

Synthetic reasoning chains. All 900 reasoning chain pairs per condition were generated by Claude Haiku. Systematic biases in Haiku's causal reasoning traces — particularly in how it structures harm chains for the intrinsic condition — may be present and are difficult to characterize without human evaluation of a stratified sample.

No adversarial robustness evaluation against the reasoning structure. The paper's jailbreak evaluation uses HarmBench standard prompts but does not include attacks specifically designed to challenge the causal reasoning chain — disputed-causation prompts, stakeholder-flipping scenarios, or hypothetical wrappers that engage the reasoning steps directly. Such reasoning-targeted attacks are the strongest test of the consequentialist robustness argument and are deferred to future work.

Intrinsic adapter weakness on copyright behaviors. The intrinsic adapter shows lower refusal rate on the HarmBench copyright subset (per-category $R_J = 0.60$) than the deontic adapter (0.90). Copyright is excluded from the primary safety metric (see Section 4) because its harm vector is categorically distinct from physical, cyber, and harassment harm — but the differential is itself informative about how the two reasoning structures handle harm types. We hypothesize that the intrinsic adapter's four-step causal trace does not flag copyright violation as strongly net-negative because its causal harm chain produces less salient consequences (legal/IP) compared to the consequences traced for physical or cyber harm (injury, financial loss, system compromise). The deontic adapter, citing rules without tracing consequences, treats copyright on equal rule-citation footing with other categories. This is a known weakness of consequentialist reasoning frameworks more broadly: harms whose mechanisms operate primarily through institutional or legal structures, rather than direct causal chains to identifiable affected parties, are systematically underweighted. A practical deployment of the intrinsic adapter would require either explicit copyright-protection training or a complementary rule-citation layer for harm types whose mechanisms are difficult to consequentialize.

Training data clustering and hybrid regeneration. Initial synthetic data generation produced severe topic clustering in multiple subcategories — 72% of medical edge cases were ketamine variants, 77% of harm reduction cases were opioid overdose variants, 60% of historical edge cases were Nazi/Third Reich variants, and several benign subcategories clustered around single topics. This is a known failure mode of unconstrained LLM synthetic data generation: without explicit diversity constraints, a language model samples its highest-probability response to each generation prompt, which is typically the same prototypical scenario repeated with minor surface variation. The dataset was regenerated using a hybrid approach: prompts sourced from human-curated benchmarks (HarmBench, JailbreakBench, XSTest, Dolly, OR-Bench) with SBERT-based deduplication (paraphrase-MiniLM-L6-v2, threshold=0.85), and only reasoning chains generated via Haiku. This approach guarantees prompt-level diversity through source dataset design and embedding-similarity validation, while limiting generation to the one component (structured reasoning traces) for which no open-source substitute exists.

8 Conclusion

This paper reports a controlled comparison of two safety-reasoning training paradigms — consequentialist causal-consequence-modeling and deontic rule-citation — at LoRA supervised fine-tuning scale on Mistral-7B-Instruct-v0.3. Under matched architecture, training data, optimizer, and adapter rank, both paradigms produce near-perfect jailbreak resistance ($R_J = 1.000$ vs. 0.983, statistically equivalent) with no statistically detectable cross-domain capability differential between any pair of configurations on either MMLU or HumanEval probes. We find a modest but directionally consistent cost asymmetry favoring the consequentialist paradigm on three subjective-task dimensions: benign-task utility (intrinsic 0.696 vs. deontic 0.602; intrinsic loses 2% of baseline utility, deontic loses 15%), violation severity ($\bar{V}$ 0.000 vs. 0.012), and over-refusal on benign tasks (0/24 vs. 1/24). The directional consistency across all three dimensions is the substantive finding; individual magnitudes are individually modest. We additionally characterize a template-dominance failure mode shared by both adapters: under explicit instruction to bypass their trained reasoning templates, both adapters override the instruction by producing extensive structured content and default to letter "A" at high rates, with different mechanisms (filter producing full structured commentary; intrinsic applying causal-trace template to the meta-question of whether to answer) but comparable behavioral outcomes.

The originally hypothesized capability-displacement asymmetry — that one paradigm would actively suppress general capability while the other would not — is not detectable at SFT scale. Both ablations recover baseline-equivalent capability cleanly. We position this negative finding as consistent with the trajectory framing's prediction: SFT, being the most surgically separable training intervention, is the level at which structural integration of alignment into reasoning is least likely to manifest. The purpose of the SFT-level experiment is to establish the baseline against which deeper-intervention experiments will be compared. The companion DPO experiment (Exp03), in progress, applies the same training-data construction and evaluation protocol to preference-optimization fine-tuning under matched conditions; the planned PPO experiment (Exp04) extends to reward-driven RL fine-tuning. The trajectory hypothesis predicts that as training intervention depth increases, the structural separability of any alignment mechanism should decrease, and load-bearing structural integration — if it exists — should become detectable at deeper levels. The current paper's contribution is to report what the SFT-level evidence does and does not show, propose the inverted ablation diagnostic as the comparison instrument for the trajectory experiments, and characterize the modest cost asymmetry that is visible at this depth.

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